

Class 11 Biology: High-Yield Diagram & Mark Breakdown

Chapter 4: Animal Kingdom (NCERT Page-wise Priority Guide)

Instructions for Students: Practice schematic representations of body plans (symmetry, germ layers, coelom) and structural diagrams neatly with pencil. Ensure all identifying features and diagnostic phylum characteristics are labelled in CAPITAL letters.

Fig No.	Diagram / Topic	NCERT	Marks	Priority	Key Labelling Points / Diagnostic Features
Fig 4.16	Chordata Fundamental Characteristics	Pg 46	3 M	Top Priority	Dorsal hollow nerve cord, Notochord, Pharyngeal gill slits, Post-anal part / tail
Fig 4.3 (a–c)	Body Cavity (Coelom Types)	Pg 40	3 M	Top Priority	(a) Coelomate (mesodermal lining), (b) Pseudocoelomate (scattered pouches in Aschelminthes), (c) Acoelomate (Platyhelminthes)
Fig 4.2 (a, b)	Germinal Layers (Diploblastic vs Triploblastic)	Pg 39	2 – 3 M	Top Priority	Diploblastic: Ectoderm, Endoderm, undifferentiated <i>Mesoglea</i> . Triploblastic: Ectoderm, <i>Mesoderm</i> , Endoderm
Fig 4.15	<i>Balanoglossus</i> (Hemichordata)	Pg 45	2 – 3 M	Top Priority	Proboscis (excretory proboscis gland), Collar, Long cylindrical Trunk
Fig 4.6 (a, b)	Cnidaria Body Forms (Polyp vs Medusa)	Pg 41	2 – 3 M	Top Priority	Polyp (<i>Adamsia</i> /Hydra - sessile & cylindrical), Medusa (<i>Aurelia</i> - umbrella-shaped & free swimming), Metagenesis
Fig 4.7	Cnidoblast / Cnidocyte	Pg 41	2 M	High Priority	Stinging capsule / Nematocyst, Thread tube, Operculum (anchorage, defense, prey capture)
Fig 4.1 (a, b)	Body Symmetry (Radial vs Bilateral)	Pg 39	2 M	High Priority	Radial symmetry (multiple central planes divide body), Bilateral symmetry (single plane divides left & right)
Fig 4.5 & 4.10	Representative Invertebrates (Porifera & Arthropoda)	Pg 40, 43	2 M	Moderate	<i>Sycon</i> / <i>Spongilla</i> (Ostia, Spongocoel, Osculum); Arthropods (Head, Thorax, Abdomen, Jointed appendages)

KEY BOARD EXAM DRAWING & CONCEPTUAL GUIDELINES

- **Chordate Characteristics (3 Marks, Fig 4.16):** A direct drawing question in exams. Clearly show the four diagnostic features: Dorsal hollow nerve cord, Notochord, Pharyngeal gill slits, and Post-anal tail.
- **Coelom Schematics (3 Marks, Fig 4.3):** Practice drawing cross-sections depicting the distinct presence or absence of mesoderm lining and scattered pouches for pseudocoelom.
- **Table 4.1 Comparison (Pg 46):** Memorize differences between Chordates and Non-chordates (Central nervous system dorsal/ventral, Gill slits, Heart position) alongside the diagram.